CSE321 Quiz 02
Time 20 minutes

Name **ID** **Sec**

Q1. You have a File System where there are a total 1100 blocks of data region [where data can be stored]. The Inode Structure you are implementing has 4 direct blocks, 3 single indirect blocks and 1 double indirect block. The Address space needed for each block is 8 bytes. The size of the block is 256 bytes. 6

a. What will be the maximum size of the file in this system given the requirements.
b. You are trying to store a file with the size of 1036. How much will the internal fragmentation cost
c. Somehow, 100 blocks are occupied by system storage and 1000 blocks remain as data blocks. What will be the potential issue here if any.

Q2. You are implementing a file system where you are observing that you have lots of empty data blocks but still we can't store any new files. What can be the reason for it? 1

Q3. Consider these two INODE Structure: 3

INODE 1: 3 direct, 2 single indirect, 1 double indirect block.
INODE 2: 2 direct, 2 double indirect block
Data block size = 128 bytes, data block address space 8 bytes. You are storing a file of size 4352 bytes in both versions. You want to access the last byte of the file. Which implementation will ensure that the least amount of disk access count (Disk Read) in this scenario. Justify with mathematical demonstration.